CAPSTONE PROJECT REPORT

MUTUAL FUND ANALYTICS

An End-to-End Data Analytics Platform for
Fund Performance, Risk & Investor Behaviour Analysis

Capstone Project I

ABSTRACT

The Indian mutual fund industry has grown rapidly over the last decade, driven by rising retail participation, SIP-based investing, and digital access to financial products. This growth generates large volumes of data around fund performance, investor transactions, portfolio holdings, and market benchmarks — data that, if analyzed systematically, can offer meaningful insight into fund quality, investor behaviour, and portfolio risk.

This Capstone Project, “Mutual Fund Analytics”, presents an end-to-end analytics solution built around a simulated but realistic mutual fund dataset comprising 40 schemes across 10 fund houses, along with NAV history, AUM, SIP inflows, investor transactions, portfolio holdings, and benchmark index data. The project follows a structured data pipeline: raw CSV datasets are ingested and validated, cleaned using Pandas, and loaded into a SQLite star-schema database consisting of dimension and fact tables. Jupyter notebooks are used to perform exploratory data analysis, compute performance metrics (CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown), and carry out advanced analytics including rolling Sharpe ratios, Historical Value-at-Risk (VaR) and Conditional VaR, sector concentration (Herfindahl-Hirschman Index), investor cohort analysis, SIP continuity/at-risk investor flagging, tracking error against benchmarks, and a simple risk-based fund recommender.

The processed data and computed metrics are consolidated into an interactive Power BI dashboard spanning industry overview, fund performance, investor analytics, SIP/market trends, and NAV-level detail. Key findings include a strong concentration of AUM among a handful of large fund houses, a majority of schemes falling in the Moderate-to-High risk band, meaningful dispersion in risk-adjusted returns (Sharpe Ratio) across similar-category funds, and a sizeable proportion of SIP investors showing irregular contribution patterns that flag them as at-risk of discontinuation.

The project demonstrates the practical application of the complete data analytics lifecycle — ingestion, cleaning, storage, computation, visualization, and insight generation — to a real-world financial domain, and provides a reusable foundation that can be extended with live market data, machine learning-based recommendation, and automated reporting.

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1. Introduction

The mutual fund industry acts as a bridge between individual investors and capital markets, pooling money from thousands of retail and institutional participants into professionally managed portfolios of equity, debt, and hybrid instruments. In India, the industry has expanded rapidly over the past decade on the back of increasing financial literacy, digital onboarding (KYC, UPI-based SIPs), and consistent promotion of Systematic Investment Plans (SIPs) as a disciplined investment vehicle.

With this growth comes a large and continuously expanding volume of data: daily Net Asset Values (NAV), fund-level performance figures, Assets Under Management (AUM), investor transaction logs, portfolio holdings, and benchmark index movements. Extracting actionable insight from this data requires a structured analytics pipeline — one that can ingest raw data, clean and validate it, store it efficiently, compute domain-specific financial metrics, and present the results in a form that is usable by fund analysts, distributors, or investors themselves.

This project, Mutual Fund Analytics, was undertaken as Capstone Project I to build exactly such a pipeline, end-to-end, using a realistic (though anonymized/simulated) mutual fund dataset covering 40 schemes from 10 Indian fund houses. The project spans the full data analytics lifecycle: data ingestion and validation, cleaning and transformation, star-schema database design in SQLite, exploratory data analysis, performance and risk analytics, advanced analytics (VaR/CVaR, rolling Sharpe, sector concentration, investor cohorts, SIP continuity, tracking error, a simple recommender), and finally an interactive Power BI dashboard for business-facing consumption.

The report that follows documents the problem being solved, the objectives of the project, the datasets used, the technology stack, the architecture of the solution, and a section-by-section walkthrough of each stage of the pipeline, supported by screenshots, charts, and the key insights derived from the analysis.

2. Problem Statement

Retail investors and fund distributors are frequently presented with raw performance figures — 1-year, 3-year, and 5-year returns — without adequate context on the risk taken to achieve those returns, the concentration of the underlying portfolio, or how consistent an investor's own contribution behaviour has been. Fund house-level data, investor transaction data, and portfolio holding data typically reside in disconnected, unstructured CSV extracts that are not directly queryable or comparable across schemes.

This creates several concrete problems that the project sets out to address:

- Fragmented data: fund master details, NAV history, AUM, SIP inflows, transactions, and portfolio holdings arrive as ten separate raw files with no common relational structure.
- Lack of risk-adjusted comparison: raw returns alone do not reveal whether a fund's performance was achieved with excessive volatility, concentration, or drawdown risk.
- No systematic view of investor behaviour: SIP discontinuation, irregular contribution gaps, and cohort-wise investment patterns are not visible from raw transaction logs.
- Absence of a consolidated reporting layer: there is no dashboard that lets a stakeholder explore fund performance, investor analytics, and market trends together in one place.

The project therefore aims to design and implement a complete pipeline — from raw data to a queryable database to computed analytics to an interactive dashboard — that resolves these gaps for a mutual fund dataset.

3. Objectives

The specific objectives defined for this Capstone Project are as follows:

- To ingest and validate ten raw mutual fund datasets (fund master, NAV history, AUM, SIP inflows, category inflows, folio counts, scheme performance, investor transactions, portfolio holdings, and benchmark indices).
- To clean and standardize the datasets — handling missing values, duplicate records, inconsistent data types, and invalid entries — and persist the cleaned outputs for downstream use.
- To design and populate a normalized SQLite star-schema database consisting of a fund dimension table and four fact tables (NAV, transactions, performance, AUM).
- To perform exploratory data analysis (EDA) on fund categories, risk categories, fund house distribution, NAV trends, and transaction patterns.
- To compute standard performance and risk metrics — Daily Returns, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown — and rank funds using a composite Fund Scorecard.
- To carry out advanced analytics: Historical VaR & CVaR, 90-day Rolling Sharpe Ratio, Sector HHI concentration, Investor Cohort Analysis, SIP Continuity / At-Risk investor flagging, Tracking Error vs. benchmarks, and a simple risk-based Fund Recommender.
- To build an interactive Power BI dashboard consolidating industry overview, fund performance, investor analytics, SIP/market trends, and NAV-level detail.
- To document the entire pipeline, its architecture, and the resulting insights in a structured project report.

4. Dataset Description

The project uses the Bluestock Mutual Fund dataset, a structured collection of ten CSV files simulating real-world mutual fund industry data. Together, the files cover fund master data, price history, capital flows, investor transactions, portfolio composition, and market benchmarks.

#	Dataset File	Key Fields	Description
1	01_fund_master.csv	amfi_code, fund_house, scheme_name, category, sub_category, risk_category	Master reference data for all 40 mutual fund schemes across 10 fund houses
2	02_nav_history.csv	amfi_code, nav_date, nav	Historical daily Net Asset Value records used for return and risk calculations
3	03_aum_by_fund_house.csv	fund_house, aum_crore, num_schemes	Assets Under Management by fund house across reporting periods
4	04_monthly_sip_inflows.csv	month, sip_amount_crore	Industry-wide monthly SIP inflow trend data
5	05_category_inflows.csv	category, inflow_crore	Net inflows by fund category (Equity / Debt)
6	06_industry_folio_count.csv	month, folio_count	Total investor folio counts across the industry, by month
7	07_scheme_performance.csv	return_1yr/3yr/5yr_pct, alpha, beta, sharpe, sortino, std_dev, max_drawdown, aum, expense_ratio	Pre-computed and derived performance/risk indicators per scheme
8	08_investor_transactions.csv	investor_id, transaction_date, amfi_code, transaction_type, amount_inr, state, city, age_group	32,778 individual SIP / Lumpsum / Redemption transactions with investor demographics
9	09_portfolio_holdings.csv	amfi_code, stock_name, sector, weight_pct	Underlying stock/sector holdings for each equity scheme, used for HHI concentration analysis
10	10_benchmark_indices.csv	index_name, index_date, index_value	Daily values of benchmark indices (NIFTY 50, NIFTY 100, etc.) used for Alpha/Beta and tracking error

Dataset Scale

- 40 mutual fund schemes across 10 fund houses (SBI, HDFC, ICICI Prudential, Nippon India, Kotak Mahindra, Axis, Aditya Birla Sun Life, UTI, Mirae Asset, DSP).
- 2 broad categories — Equity (34 schemes) and Debt (6 schemes) — spanning 12 sub-categories such as Large Cap, Mid Cap, Small Cap, Flexi Cap, ELSS, Liquid, and Gilt.
- 46,000 NAV history records used for return, volatility, drawdown, and Sharpe/Sortino computation.
- 32,778 investor transaction records spanning SIP, Lumpsum, and Redemption transaction types.
- A supplementary live_nav folder containing NAV data fetched from the public mfapi.in API for six benchmark large-cap schemes, used to validate the static dataset against a live source.

A complete field-level data dictionary (column names, data types, and business definitions for every dataset) was prepared and is maintained in reports/data_dictionary.md.

5. Technology Stack

The project uses a Python-centric, open-source analytics stack, chosen for its strong ecosystem support for data manipulation, statistical computation, and visualization, paired with SQLite for lightweight relational storage and Power BI for business-facing dashboarding.

Layer	Technology	Purpose
Language	Python 3.11	Core scripting for ingestion, cleaning, and analytics
Data Processing	Pandas, NumPy	Data wrangling, cleaning, numerical computation
Statistics	SciPy	Statistical calculations supporting risk metrics
Visualization	Matplotlib, Seaborn, Plotly	EDA charts, performance charts, rolling metrics plots
Database	SQLite + SQLAlchemy	Lightweight relational star-schema database and ORM-style loading
Notebooks	Jupyter Notebook	Interactive, documented analysis (ingestion, cleaning, EDA, performance, advanced analytics)
Dashboard / BI	Microsoft Power BI	Interactive business-facing dashboard across 5 report pages
Data Access	Requests + mfapi.in API	Live NAV data collection for validation
Version Control	Git & GitHub	Source control and project hosting (Madhavarao111)
Environment	VS Code, Windows + PowerShell	Primary development environment

Key Python Libraries

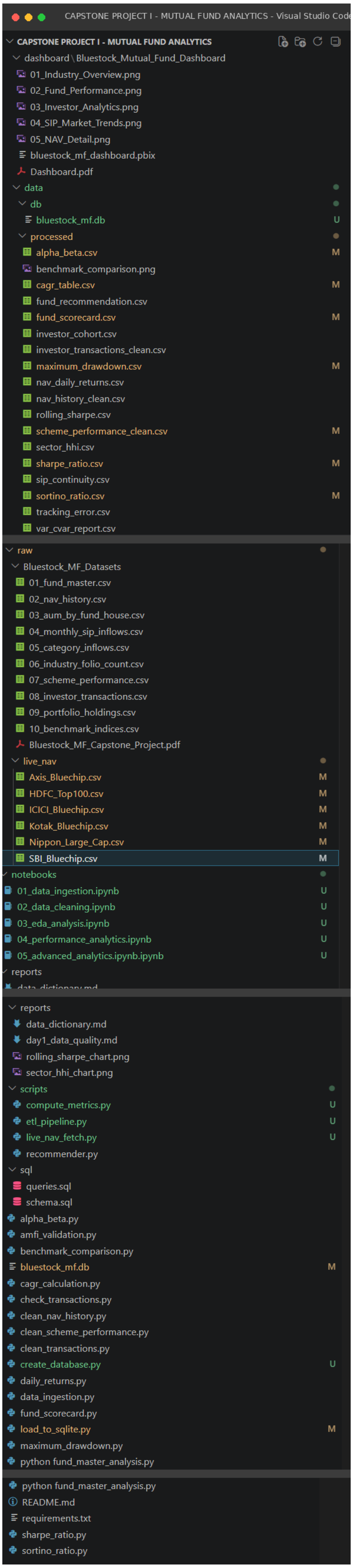
- pandas, numpy — data loading, cleaning, transformation, and vectorized financial calculations.
- matplotlib, seaborn, plotly — static and interactive chart generation for EDA and analytics notebooks.
- sqlalchemy — database engine creation and DataFrame-to-SQL loading (to_sql).
- requests — HTTP calls to the mfapi.in public NAV API for live data validation.
- scipy, jupyter — statistical routines and the interactive notebook environment (see requirements.txt).

6. Project Architecture

The solution follows a layered analytics architecture: a Raw Data layer holding the original CSV extracts, a Processing layer of standalone Python scripts and Jupyter notebooks that clean and enrich the data, a Storage layer backed by a SQLite star-schema database, an Analytics layer that computes performance and risk metrics, and a Presentation layer consisting of static report charts and an interactive Power BI dashboard.

Folder Structure

The project repository is organized as follows, separating raw data, processed outputs, database files, notebooks, standalone scripts, SQL definitions, reports, and the Power BI dashboard into dedicated folders:


CAPSTONE PROJECT I - MUTUAL FUND ANALYTICS - Visual Studio Code
CAPSTONE PROJECT I - MUTUAL FUND ANALYTICS
dashboard \ Bluestock_Mutual_Fund_Dashboard
01_Industry_Overview.png
02_Fund_Performance.png
03_Investor_Analytics.png
04_SIP_Market_Trends.png
05_NAV_Detail.png
bluestock_mf_dashboard.pbix
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data
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alpha_beta.csv M
benchmark_comparison.png
cagr_table.csv M
fund_recommendation.csv
fund_scorecard.csv M
investor_cohort.csv
investor_transactions_clean.csv
maximum_drawdown.csv M
nav_daily_returns.csv
nav_history_clean.csv
rolling_sharpe.csv
scheme_performance_clean.csv M
sector_hhi.csv
sharpe_ratio.csv M
sip_continuity.csv
sortino_ratio.csv M
tracking_error.csv
var_cvar_report.csv
raw
Bluestock_MF_Datasets
01_fund_master.csv
02_nav_history.csv
03_aum_by_fund_house.csv
04_monthly_sip_inflows.csv
05_category_inflows.csv
06_industry_folio_count.csv
07_scheme_performance.csv
08_investor_transactions.csv
09_portfolio_holdings.csv
10_benchmark_indices.csv
Bluestock_MF_Capstone_Project.pdf
live_nav
Axis_Bluechip.csv M
HDFC_Top100.csv M
ICICI_Bluechip.csv M
Kotak_Bluechip.csv M
Nippon_Large_Cap.csv M
SBI_Bluechip.csv M
notebooks
01_data_ingestion.ipynb U
02_data_cleaning.ipynb U
03_eda_analysis.ipynb U
04_performance_analytics.ipynb U
05_advanced_analytics.ipynb.ipynb U
reports
reports
data_dictionary.md
day1_data_quality.md
rolling_sharpe_chart.png
sector_hhi_chart.png
scripts
compute_metrics.py U
etl_pipeline.py U
live_nav_fetch.py U
recommender.py
sql
queries.sql
schema.sql
alpha_beta.py
amfi_validation.py
benchmark_comparison.py
bluestock_mf.db M
cagr_calculation.py
check_transactions.py
clean_nav_history.py
clean_scheme_performance.py
clean_transactions.py
create_database.py U
daily_returns.py
data_ingestion.py
fund_scorecard.py
load_to_sqlite.py M
maximum_drawdown.py
python fund_master_analysis.py
python fund_master_analysis.py
README.md
requirements.txt
sharpe_ratio.py
sortino_ratio.py

Figure 6.1: Project folder structure as organized in the VS Code Explorer

Architecture Flow

- Raw Layer: data/raw/Bluestock_MF_Datasets (10 CSVs) + data/raw/live_nav (API-fetched NAV series).
- Cleaning Layer: clean_nav_history.py, clean_transactions.py, clean_scheme_performance.py — standalone scripts orchestrated by scripts/etl_pipeline.py.
- Storage Layer: create_database.py provisions the SQLite file; load_to_sqlite.py loads cleaned DataFrames into dim_fund, fact_nav, fact_transactions, fact_performance, and fact_aum.
- Analytics Layer: daily_returns.py, cagr_calculation.py, sharpe_ratio.py, sortino_ratio.py, alpha_beta.py, maximum_drawdown.py, fund_scorecard.py, benchmark_comparison.py — orchestrated by scripts/compute_metrics.py.
- Advanced Analytics: implemented interactively in notebooks/05_advanced_analytics.ipynb.ipynb (VaR/CVaR, rolling Sharpe, HHI, cohorts, SIP continuity, recommender) and scripts/recommender.py.
- Presentation Layer: chart PNGs in reports/, and the Power BI file dashboard/Bluestock_Mutual_Fund_Dashboard/bluestock_mf_dashboard.pbix with exported page images and a PDF export.

7. ETL Pipeline

The ETL (Extract-Transform-Load) pipeline moves data from raw CSV files to a query-ready SQLite database. It is implemented as a set of independent Python scripts that can be run individually during development, and as an orchestrated sequence via scripts/etl_pipeline.py for repeatable, end-to-end execution.

Extract

Raw datasets are read directly from data/raw/Bluestock_MF_Datasets using pandas.read_csv(). The notebooks/01_data_ingestion.ipynb notebook additionally fetches live NAV history for six benchmark schemes (HDFC Top 100, SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, Kotak Bluechip) from the public mfapi.in REST API, and stores the responses as CSV files in data/raw/live_nav for cross-validation against the static dataset.

Transform

Three dedicated cleaning scripts standardize the highest-volume and highest-risk datasets before loading:

- clean_nav_history.py — parses nav_date to datetime, sorts by (amfi_code, date), forward-fills missing NAV values per fund, and removes duplicate rows.
- clean_transactions.py — parses transaction_date, standardizes transaction_type text casing (Sip / Lumpsum / Redemption), and validates transaction amounts.
- clean_scheme_performance.py — coerces return_1yr_pct, return_3yr_pct, and return_5yr_pct to numeric types, handling any non-numeric artifacts from the source extract.

Load

scripts/etl_pipeline.py orchestrates the pipeline end-to-end via subprocess calls, running the three cleaning scripts followed by load_to_sqlite.py, which uses a SQLAlchemy engine to write each cleaned DataFrame into its corresponding SQLite table with to_sql(if_exists="replace"). After loading, the script cross-validates CSV row counts against the resulting database row counts to confirm a lossless load.

Running scripts/etl_pipeline.py: *clean_nav_history.py → clean_transactions.py → clean_scheme_performance.py → load_to_sqlite.py → “ETL Pipeline Completed Successfully!”*

Post-Load Verification

Verification output confirmed a complete and lossless ETL run: dim_fund (40 rows), fact_nav (46,000 rows), fact_transactions (32,778 rows), fact_performance (40 rows), and fact_aum (90 rows) — matching the source CSV row counts exactly.

8. Database Design

Cleaned data is loaded into a SQLite database (bluestock_mf.db) organized as a star schema — one central fund dimension table surrounded by four fact tables capturing NAV history, investor transactions, computed performance metrics, and AUM. This design keeps fund attributes normalized in a single place while allowing efficient aggregation across NAV, transaction, and performance facts using amfi_code as the common foreign key.

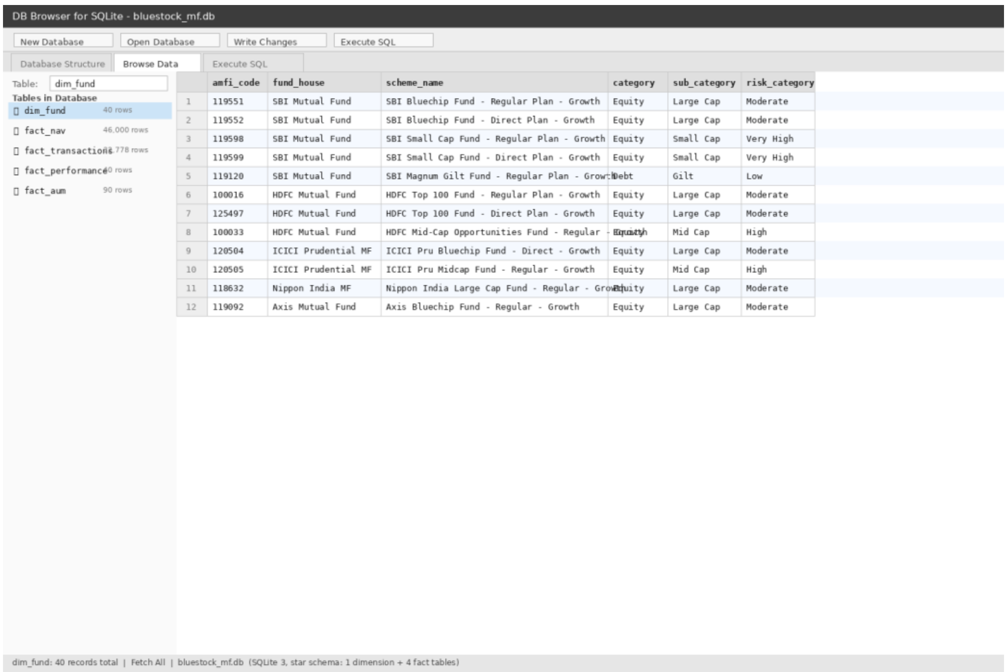
Schema Overview

Table	Type	Key Columns	Rows
dim_fund	Dimension	amfi_code (PK), fund_house, scheme_name, category, sub_category, risk_category	40
fact_nav	Fact	nav_id (PK), amfi_code (FK), nav_date, nav	46,000
fact_transactions	Fact	txn_id (PK), investor_id, amfi_code (FK), transaction_type, amount_inr, transaction_date	32,778
fact_performance	Fact	perf_id (PK), amfi_code (FK), return_1y, return_3y, return_5y	40
fact_aum	Fact	aum_id (PK), fund_house, aum_cr	90

The schema is formally defined in sql/schema.sql using standard SQL DDL with PRIMARY KEY and FOREIGN KEY constraints tying each fact table back to dim_fund.amfi_code. A library of ten analytical SQL queries — covering top funds by AUM, average monthly NAV, SIP year-over-year growth, transactions by state, low-expense funds, top funds by 1-year return and by Sharpe Ratio, average investment by transaction type, and risk/category distributions — is maintained in sql/queries.sql.

Database Browser View

The populated database was inspected using a SQLite browser to confirm table structure and row-level content. The dim_fund table below shows fund master records — AMFI code, fund house, scheme name, category, sub-category, and risk classification — for the 40 schemes loaded into the database:



	amfi_code	fund_house	scheme_name	category	sub_category	risk_category
1	119551	SBI Mutual Fund	SBI Bluechip Fund - Regular Plan - Growth	Equity	Large Cap	Moderate
2	119552	SBI Mutual Fund	SBI Bluechip Fund - Direct Plan - Growth	Equity	Large Cap	Moderate
3	119598	SBI Mutual Fund	SBI Small Cap Fund - Regular Plan - Growth	Equity	Small Cap	Very High
4	119599	SBI Mutual Fund	SBI Small Cap Fund - Direct Plan - Growth	Equity	Small Cap	Very High
5	119120	SBI Mutual Fund	SBI Magnum Gilt Fund - Regular Plan - Growth	Debt	Gilt	Low
6	100016	HDFC Mutual Fund	HDFC Top 100 Fund - Regular Plan - Growth	Equity	Large Cap	Moderate
7	125497	HDFC Mutual Fund	HDFC Top 100 Fund - Direct Plan - Growth	Equity	Large Cap	Moderate
8	100033	HDFC Mutual Fund	HDFC Mid-Cap Opportunities Fund - Regular - Growth	Equity	Mid Cap	High
9	120504	ICICI Prudential MF	ICICI Pru Bluechip Fund - Direct - Growth	Equity	Large Cap	Moderate
10	120505	ICICI Prudential MF	ICICI Pru Midcap Fund - Regular - Growth	Equity	Mid Cap	High
11	118632	Nippon India MF	Nippon India Large Cap Fund - Regular - Growth	Equity	Large Cap	Moderate
12	119092	Axis Mutual Fund	Axis Bluechip Fund - Regular - Growth	Equity	Large Cap	Moderate

Figure 8.1: bluestock_mf.db inspected in a SQLite database browser — dim_fund table

9. Data Cleaning

Data cleaning was carried out in notebooks/02_data_cleaning.ipynb and the corresponding standalone scripts, with the objective of producing analysis-ready datasets free of missing values, duplicates, and type inconsistencies. Cleaned outputs are persisted as CSV files under data/processed/ for reuse across the analytics and dashboard layers.

Cleaning Steps Applied

- NAV History: nav_date converted to a proper datetime type; records sorted by (amfi_code, date); missing NAV values forward-filled within each fund group (ffill) to preserve time-series continuity; duplicate rows dropped → saved as nav_history_clean.csv.
- Investor Transactions: transaction_date parsed to datetime; transaction_type text trimmed and title-cased for consistent grouping (Sip / Lumpsum / Redemption) → saved as investor_transactions_clean.csv.
- Scheme Performance: return_1yr_pct, return_3yr_pct, and return_5yr_pct coerced to numeric using pd.to_numeric(errors="coerce") to catch any malformed values from the source extract → saved as scheme_performance_clean.csv.
- Fund Master: validated for missing values, duplicate rows, and AMFI code integrity as part of the Day 1 data quality pass (see below).

Day 1 Data Quality Summary

An initial data quality assessment of the fund master dataset (01_fund_master.csv) reported: 40 rows across 15 columns, 10 distinct fund houses, 2 categories (Equity, Debt), 12 sub-categories, and 5 risk categories (Low to Very High). The assessment found zero missing values, zero duplicate rows, and zero missing AMFI codes — confirming the fund master dataset was clean and ready for ETL processing (see reports/day1_data_quality.md).

Overall, the cleaning stage converted all date columns to proper datetime types, forward-filled missing NAV values, removed duplicate records, and filtered invalid transaction and return entries, establishing a reliable foundation for all downstream analytics.

10. Exploratory Data Analysis

Exploratory Data Analysis was performed in notebooks/03_eda_analysis.ipynb to understand the shape, distribution, and quality of the cleaned datasets before computing derived performance metrics. The analysis covered fund category and risk distribution, fund house representation, NAV trends over time, and investor transaction-type patterns.

Fund Category & Risk Distribution

Of the 40 schemes in the dataset, 34 are Equity funds and 6 are Debt funds — confirming that Equity funds dominate the dataset. Across risk categories, 16 schemes are classified Moderate, 8 High, 6 Very High, 6 Low, and 4 Moderately High, indicating that most mutual funds in the dataset fall in the Moderate-to-High risk band.

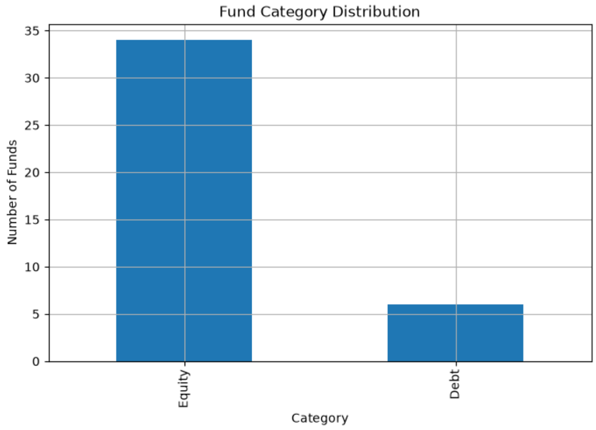


Figure 10.1: Fund category distribution — output from notebooks/03_eda_analysis.ipynb

Fund House Distribution

Multiple Asset Management Companies (AMCs) are represented in the dataset, with SBI, HDFC, ICICI Prudential, and Axis contributing several schemes each — reflecting their position as some of the largest players in the Indian mutual fund industry.

NAV Trend & Transaction Types

NAV values were observed to fluctuate day-to-day but trend upward over the long run, consistent with typical equity and hybrid fund behaviour. Among the 32,778 investor transactions, SIP transactions account for the largest share of investor activity (19,716 transactions, average ticket size ₹11,018), followed by Lumpsum (8,095 transactions, average ₹254,456) and Redemption (4,967 transactions, average ₹250,559) — confirming SIPs as the dominant mode of retail participation by transaction count, even though Lumpsum and Redemption transactions carry much larger average ticket sizes.

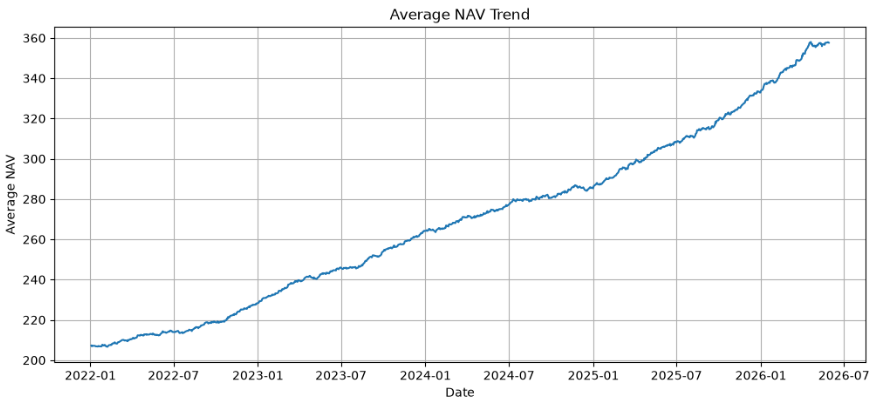


Figure 10.2: NAV trend visualization — output from notebooks/03_eda_analysis.ipynb

Top Fund Houses by AUM

SBI Mutual Fund, ICICI Prudential MF, HDFC Mutual Fund, and Nippon India MF are the largest fund houses by Assets Under Management, with average AUM of ₹943,444 crore, ₹699,222 crore, ₹636,889 crore, and ₹434,333 crore respectively across the reporting periods captured in fact_aum. Larger AUM generally reflects greater investor trust and a broader asset base.

11. Performance Analytics

notebooks/04_performance_analytics.ipynb.ipynb computes the core suite of return and risk-adjusted performance metrics used throughout the rest of the project: Daily Returns, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and a composite Fund Scorecard, followed by a Tracking Error and Benchmark Comparison analysis.

Metrics Computed

- Daily Returns (daily_returns.py) — percentage change in NAV day-over-day, the base series for all downstream risk metrics.
- CAGR (cagr_calculation.py) — Compound Annual Growth Rate over the available NAV history, saved to cagr_table.csv.
- Sharpe Ratio (sharpe_ratio.py) — excess return per unit of total volatility (standard deviation of returns).
- Sortino Ratio (sortino_ratio.py) — excess return per unit of downside deviation, penalizing only harmful volatility.
- Alpha & Beta (alpha_beta.py) — excess return over the benchmark (Alpha) and sensitivity to benchmark movement (Beta), computed against relevant NIFTY indices.
- Maximum Drawdown (maximum_drawdown.py) — the largest peak-to-trough decline in NAV, indicating worst-case historical loss.
- Fund Scorecard (fund_scorecard.py) — a composite ranking combining CAGR, Sharpe Ratio, Alpha, Expense Ratio, and Maximum Drawdown ranks into a single Score_100.

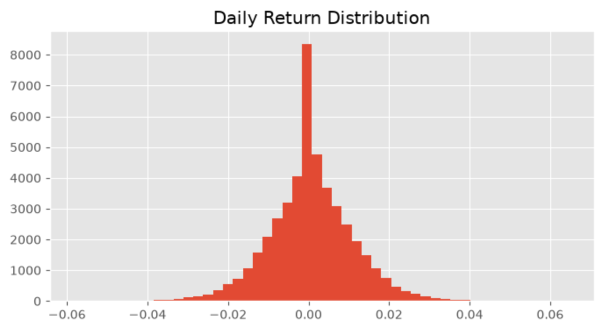


Figure 11.1: Sample performance metric output — notebooks/04_performance_analytics.ipynb.ipynb

Fund Scorecard — Top Ranked Funds

The composite Fund Scorecard, which blends CAGR, Sharpe Ratio, Alpha, expense ratio, and drawdown ranks, placed Mirae Asset Large Cap Fund at the top (Score 100.00, 34.00% 3Y CAGR, Sharpe 1.4483), followed by ICICI Pru Midcap Fund (Score 94.44) and Kotak Flexicap Fund (Score 94.09). This scorecard gives a single, balanced ranking rather than relying on raw returns alone, which can be misleading when risk levels differ across funds.

Benchmark Comparison

Fund NAV growth was plotted against relevant benchmark indices (e.g. NIFTY 50 / NIFTY 100) to visually assess whether each scheme outperformed or underperformed its passive benchmark over the observed period:

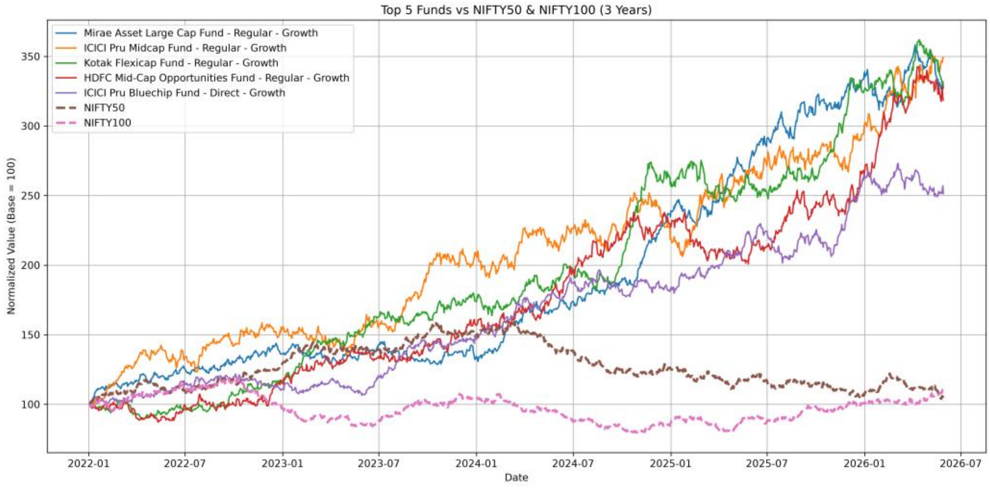


Figure 11.2: Benchmark Comparison — fund NAV growth vs. benchmark index (benchmark_comparison.py)

12. Advanced Analytics

Beyond standard performance metrics, notebooks/05_advanced_analytics.ipynb.ipynb implements six advanced analyses that provide a deeper, risk- and behaviour-aware view of the fund universe and its investors.

12.1 Historical VaR & CVaR

Historical Value-at-Risk (95%) and Conditional VaR were computed for all 40 schemes from their daily return series. VaR estimates the maximum expected one-day loss with 95% confidence, while CVaR measures the average loss during the worst 5% of trading days. As expected, Small Cap schemes dominate the highest-risk list:

Scheme	VaR (95%)	CVaR (95%)
SBI Small Cap Fund – Direct Plan	-2.69%	-3.24%
Axis Small Cap Fund – Regular	-2.62%	-3.17%
ABSL Small Cap Fund – Regular	-2.60%	-3.25%
Nippon India Small Cap Fund – Regular	-2.54%	-3.23%
SBI Small Cap Fund – Regular Plan	-2.45%	-3.06%

Funds with larger negative VaR/CVaR values exhibit higher downside risk, reinforcing that small-cap equity exposure carries materially greater tail risk than large-cap or debt exposure. The chart below (notebook output) visualizes the ten highest-risk schemes by 95% VaR:

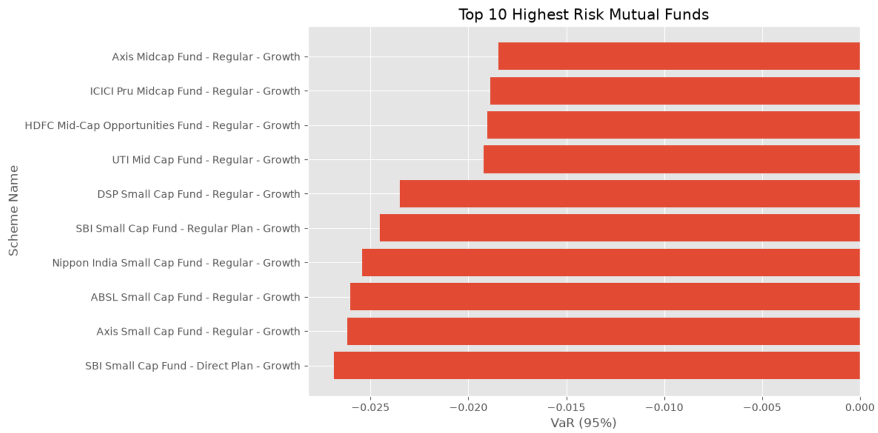


Figure 12.0: Top 10 highest-risk mutual funds by VaR (95%) — notebooks/05_advanced_analytics.ipynb.ipynb

12.2 Rolling 90-Day Sharpe Ratio

A rolling 90-day Sharpe Ratio was calculated for each scheme using a moving window of mean return and standard deviation, and plotted over time for five key funds to visualize how risk-adjusted performance evolves rather than relying on a single static Sharpe value:

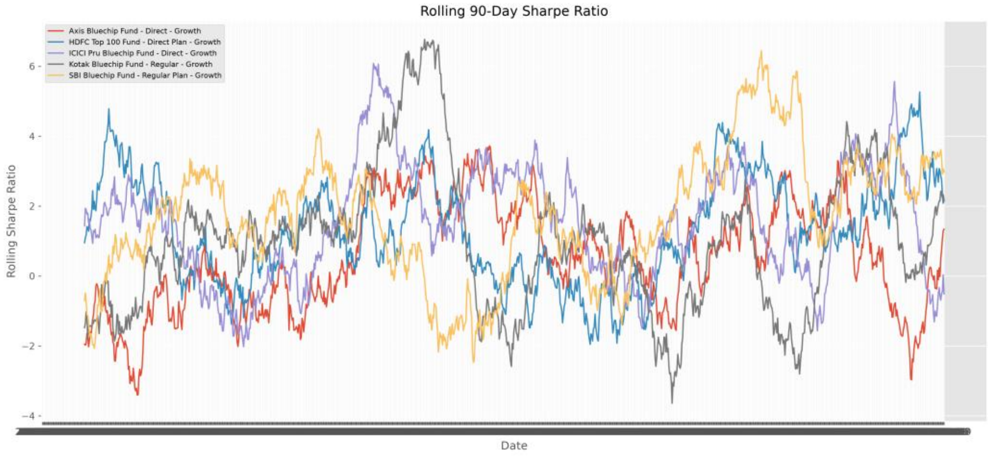


Figure 12.1: Rolling 90-Day Sharpe Ratio for five key funds (reports/rolling_sharpe_chart.png)

A higher rolling Sharpe Ratio indicates better risk-adjusted returns for that window; the chart reveals periods of relative outperformance and underperformance across funds that a single point-in-time Sharpe figure would mask.

12.3 Investor Cohort Analysis

Investors were grouped into cohorts based on the year of their first transaction. For each cohort, average SIP amount, total investment, and top fund preference were computed — enabling comparison of how investment behaviour and fund choice differ between investors who joined in different years.

12.4 SIP Continuity / At-Risk Investor Analysis

For investors with six or more SIP transactions, the average gap between consecutive SIP dates was calculated. Investors with an average gap greater than 35 days were flagged as “At Risk” of SIP discontinuation, while those at or under 35 days were classified “Regular.” Of the qualifying investor base, 1,332 investors were flagged At Risk against only 30 classified Regular — a striking signal that the large majority of frequent SIP investors in this dataset show irregular contribution patterns, which would warrant proactive retention outreach in a real fund house setting.

12.5 Simple Fund Recommender

scripts/recommender.py and the corresponding notebook section implement a rule-based recommender that accepts a user's risk appetite (Low, Moderate, or High), filters the fund universe by matching risk_category, and ranks the filtered funds by Sharpe Ratio in descending order — returning the top 3 recommended schemes for that risk profile.

12.6 Sector HHI Concentration

The Herfindahl-Hirschman Index ($HHI = \Sigma\ weight^2$) was computed for each equity scheme from its portfolio holdings to quantify sector/stock concentration. Higher HHI indicates a more concentrated portfolio; lower HHI indicates broader diversification. Axis Bluechip Fund (HHI 0.206) and ABSL Small Cap Fund (HHI 0.201) were identified as the most concentrated (“Highly Concentrated”) equity schemes in the dataset, while several others fell in the “Moderately Concentrated” band:

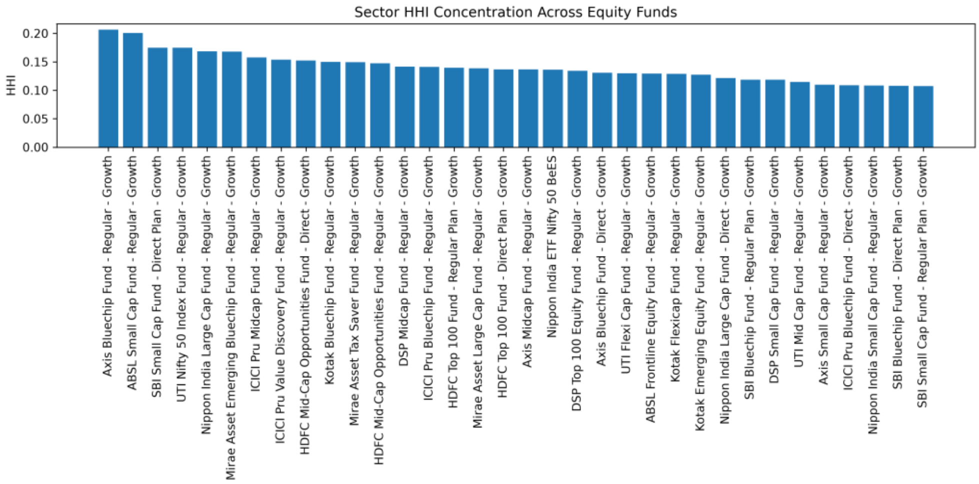


Figure 12.2: Sector HHI concentration across equity schemes (reports/sector_hhi_chart.png)

Comparing HHI across equity funds helps identify schemes that carry concentration risk despite otherwise attractive headline returns.

13. Dashboard

All processed and computed datasets — fund master, NAV, AUM, SIP inflows, transactions, and derived performance/risk metrics — were consolidated into an interactive Power BI dashboard (dashboard/Bluestock_Mutual_Fund_Dashboard/bluestock_mf_dashboard.pbix), exported as five report pages plus a combined PDF, giving business-facing stakeholders a single place to explore the mutual fund industry data.

13.1 Industry Overview

A high-level view of AUM by fund house, fund count by category, and overall industry scale.

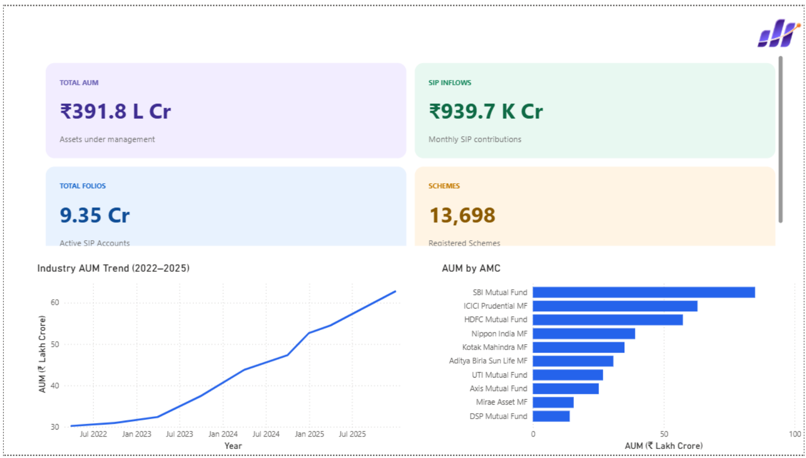


Figure 13.1: Power BI Dashboard — Industry Overview page

13.2 Fund Performance

Scheme-level performance comparison including returns, Sharpe/Sortino ratios, and the Fund Scorecard ranking.

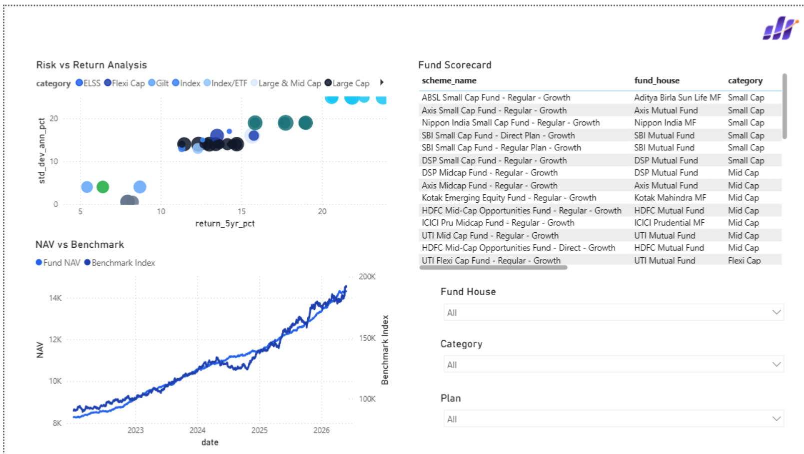


Figure 13.2: Power BI Dashboard — Fund Performance page

13.3 Investor Analytics

Investor transaction patterns by state, city tier, age group, and payment mode, alongside SIP continuity / at-risk investor flags.

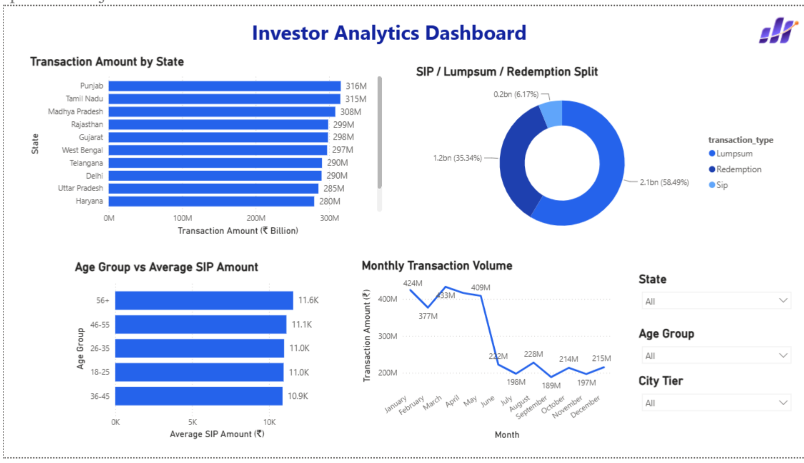


Figure 13.3: Power BI Dashboard — Investor Analytics page

13.4 SIP & Market Trends

Monthly SIP inflow trends, category-wise inflows, and folio growth across the industry.

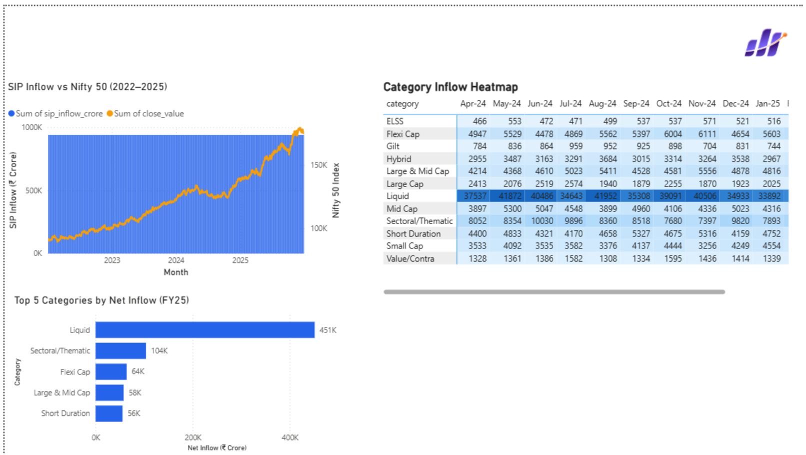


Figure 13.4: Power BI Dashboard — SIP & Market Trends page

13.5 NAV Detail

Drill-down NAV history view for individual schemes, supporting fund-level trend inspection.

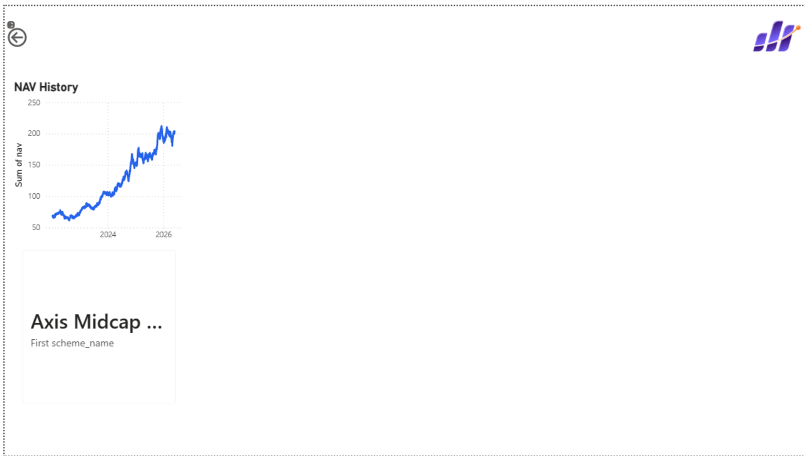


Figure 13.5: Power BI Dashboard — NAV Detail page

14. Results & Insights

Bringing together the EDA, performance analytics, and advanced analytics stages, the following key results and insights emerged from the analysis:

- AUM is concentrated among a small number of large fund houses — SBI, ICICI Prudential, HDFC, and Nippon India together account for the majority of average AUM across the dataset, consistent with real-world industry concentration.
- The scheme universe skews Equity-heavy (34 of 40 schemes) and Moderate-to-High risk (28 of 40 schemes across Moderate/High/Very High), reflecting a growth-oriented fund lineup.
- Risk-adjusted ranking changes the leaderboard: the composite Fund Scorecard placed Mirae Asset Large Cap Fund (34.00% CAGR, Sharpe 1.4483) ahead of funds with similar or even higher raw CAGR but weaker Sharpe, Alpha, or drawdown profiles — demonstrating why raw return alone is an incomplete performance measure.
- Small Cap schemes carry materially higher tail risk, occupying the top of both the VaR/CVaR list and, generally, higher expected volatility — useful context for risk-averse investors.
- SIP is the dominant transaction mode by count (19,716 of 32,778 transactions, ~60%) though Lumpsum and Redemption transactions carry roughly 23x larger average ticket sizes (~₹250K+ vs. ~₹11K).
- A large majority of frequent SIP investors (1,332 of 1,362, ~98%) show an average inter-SIP gap exceeding 35 days, flagging them as “At Risk” of discontinuation — a strong signal for retention-focused outreach in practice.
- Portfolio concentration (HHI) varies meaningfully even within similarly categorized equity funds, with Axis Bluechip Fund and ABSL Small Cap Fund showing the highest concentration — a risk dimension not visible from returns or Sharpe Ratio alone.
- The rolling 90-day Sharpe Ratio reveals that risk-adjusted performance is not static — funds move in and out of favourable risk-adjusted return regimes over time, reinforcing the value of monitoring performance over rolling windows rather than single static snapshots.

15. Conclusion

This Capstone Project successfully implemented a complete, end-to-end mutual fund analytics pipeline — from raw CSV ingestion through cleaning, star-schema database design, exploratory analysis, standard and advanced performance/risk analytics, and finally an interactive Power BI dashboard. The pipeline was validated at each stage: data quality checks confirmed clean source data, ETL verification confirmed a lossless load of 46,000+ NAV records and 32,778+ transactions into SQLite, and the computed metrics (Sharpe, Sortino, Alpha, Beta, Maximum Drawdown, VaR/CVaR, HHI) were cross-checked against domain expectations (e.g., small-cap funds showing the highest tail risk, as expected).

The project demonstrates practical, applied competence across the full data analytics lifecycle in a real-world financial domain: Python-based data engineering, relational database design, statistical and financial computation, and business intelligence visualization. Beyond the technical pipeline, the analysis surfaced genuinely actionable insights — particularly around AUM concentration, risk-adjusted fund ranking, small-cap tail risk, and the high proportion of SIP investors at risk of discontinuation — that would be directly useful to a fund house, distributor, or investor in practice.

Overall, the Mutual Fund Analytics project meets its stated objectives and provides a solid, extensible foundation for further work, as outlined in the Future Scope section that follows.

16. Future Scope

While the current implementation meets the objectives of Capstone Project I, several extensions would increase its real-world applicability and analytical depth:

- Live data integration: extend the live_nav API integration (currently limited to six benchmark schemes) to cover all 40 schemes on a scheduled basis, keeping the database continuously up to date.
- Machine learning-based recommendation: replace the current rule-based recommender with a model that incorporates investor risk profile, investment horizon, past behaviour, and fund similarity, rather than a single-metric (Sharpe Ratio) ranking.
- Predictive SIP-discontinuation modeling: use the SIP continuity / at-risk flag as a labeled target to train a classification model that predicts discontinuation risk earlier, from fewer transactions.
- Migration to a production-grade database: move from SQLite to PostgreSQL or a cloud data warehouse (e.g., BigQuery, Snowflake) to support larger data volumes and concurrent access.
- Automated pipeline orchestration: convert the current sequential script-based ETL into a scheduled, monitored pipeline using a tool such as Apache Airflow or Prefect.
- Deeper portfolio analytics: extend the sector HHI work to full portfolio overlap analysis across funds, and add factor-based attribution (value, growth, momentum) beyond Alpha/Beta.
- Web-based self-service dashboard: complement the Power BI dashboard with a lightweight web application (e.g., Streamlit or Flask) allowing investors to query the recommender and view fund analytics without a Power BI license.

17. References

[1] AMFI (Association of Mutual Funds in India) — Official mutual fund industry data and NAV disclosure standards. https://www.amfiindia.com

[2] mfapi.in — Public REST API used for live NAV history retrieval during data ingestion. https://www.mfapi.in

[3] Bluestock — Bluestock Mutual Fund Datasets (Capstone dataset source), data/raw/Bluestock_MF_Datasets/Bluestock_MF_Capstone_Project.pdf

[4] McKinley, S. — Sharpe Ratio and Sortino Ratio methodology, standard financial risk-adjusted return formulas.

[5] Herfindahl, O. C. & Hirschman, A. O. — Herfindahl-Hirschman Index (HHI), concentration measurement methodology.

[6] Pandas Development Team — pandas documentation. https://pandas.pydata.org/docs/

[7] SQLAlchemy Documentation — https://docs.sqlalchemy.org/

[8] Microsoft — Power BI Documentation. https://learn.microsoft.com/power-bi/

[9] Matplotlib, Seaborn, Plotly — official documentation for chart generation used across notebooks.

[10] Python Software Foundation — Python 3.11 Documentation. https://docs.python.org/3.11/